Supply Chain Optimization

Forecasting

Forecasting approaches

- Extrapolation of historical data
- Forecasts based on other factors (sales campaign, dependent items, economic conditions, weather forecast, expert knowledge)

Constant model

$$x_t = a + \varepsilon_t$$

x_t = demand in period t,
a = average demand per period,
ε_τ = independent random deviation with mean zero.

Trend model

$$x_t = a + bt + \varepsilon_t$$

x_t = demand in period t,
a = average demand in period 0,
b = trend, that is the systematic increase or decrease per period,
ε_τ = independent random deviation with mean zero.

Trend-seasonal model

$$x_t = (a + bt)F_t + \varepsilon_t$$

x_t = demand in period t,
a = average demand in period 0,
b = trend, that is the systematic increase or decrease per period,
F_t = seasonal index in period t,
ε_τ = independent random deviation with mean zero.

$$\sum_{k=1}^{T} F_{t+k} = T$$

Estimating model parameters

Moving average

$$\hat{x}_{t,\tau} = \hat{a}_t = (x_t + x_{t-1} + x_{t-2} + \ \ldots \ + x_{t-N+1}) / N$$

$\hat{a}_t$ = estimate of a after observing the demand in period t,

$\hat{x}_{t,\tau}$ = forecast for period $\tau > t$ after observing the demand in period t.

Exponential smoothing

$$\hat{x}_{t,\tau} = \hat{a}_t = (1-\alpha)\hat{a}_{t-1} + \alpha x_t$$

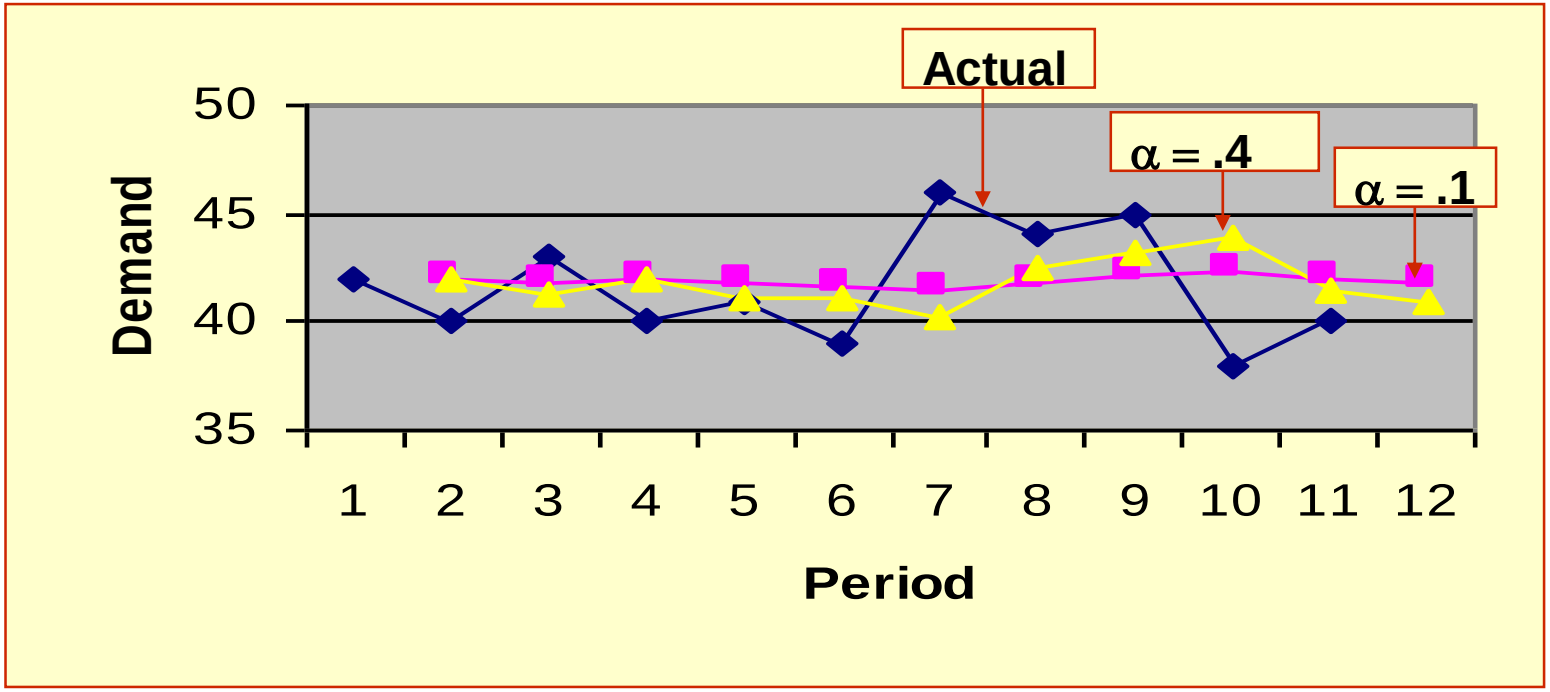
α = smoothing constant ($0 < \alpha < 1$).

$$\hat{a}_t = (1-\alpha)\hat{a}_{t-1} + \alpha x_t = (1-\alpha)((1-\alpha)\hat{a}_{t-2} + \alpha x_{t-1}) + \alpha x_t$$

$$= \alpha x_t + \alpha(1-\alpha)x_{t-1} + (1-\alpha)^2 \hat{a}_{t-2} = \ldots = \alpha x_t + \alpha(1-\alpha)x_{t-1}$$

$$+\alpha(1-\alpha)^2 x_{t-2} + \ \ldots \ + \alpha(1-\alpha)^n x_{t-n} + (1-\alpha)^{n+1}\hat{a}_{t-n-1}$$

Picking a Smoothing Constant



Minimize forecast errors,
discussed later

Exponential smoothing with trend

first suggested by Holt (1957)

$$\hat{x}_{t,t+k} = \hat{a}_t + k \cdot \hat{b}_t$$

$$\hat{a}_t = (1-\alpha)(\hat{a}_{t-1} + \hat{b}_{t-1}) + \alpha x_t$$

$$\hat{b}_t = (1-\beta)\hat{b}_{t-1} + \beta(\hat{a}_t - \hat{a}_{t-1})$$

Winters' trend-seasonal method

$$\hat{x}_{t,t+k} = (\hat{a}_t + k \cdot \hat{b}_t)\hat{F}_{t+k}$$

$$\hat{a}_t = (1-\alpha)(\hat{a}_{t-1} + \hat{b}_{t-1}) + \alpha(x_t / \hat{F}_t) \qquad \hat{b}_t = (1-\beta)\hat{b}_{t-1} + \beta(\hat{a}_t - \hat{a}_{t-1})$$

seasonal indices

$$\hat{F}'_t = (1-\gamma)\hat{F}_t + \gamma(x_t / \hat{a}_t)$$

$$\hat{F}'_{t-i} = \hat{F}_{t-i} \quad \text{and} \quad \hat{F}_{t-i} = \hat{F}'_{t-i}(T / \sum_{k=0}^{T-1} \hat{F}'_{t-k}) \qquad \text{for } i = 1, 2, \ldots, T-1$$

also applied to future periods until the indices are updated again

$$\hat{F}_{t-i+kT} = \hat{F}_{t-i}$$

for i = 0, 1, ... , T-1,
and k = 1, 2, ...

Initialisation

Constant

Constant

$$x_t = a + \varepsilon_t$$

Moving average

$$\hat{a}_{-1} = \frac{x_{-1} + x_{-2} + x_{-3} + \ldots + x_{-N}}{N}$$

Exponentially weighted Moving average

$$\hat{a}_{-1} = \frac{x_{-1} + (1-\alpha)x_{-2} + (1-\alpha)^2 x_{-3} + \ldots + (1-\alpha)^{N-1} x_{-N}}{1 + (1-\alpha) + (1-\alpha)^2 + \ldots + (1-\alpha)^{N-1}}$$

Trend

$$x_t = a + bt + \varepsilon_t$$

Regression (ordinary least squares)

$$\hat{b}_{-1} = \frac{\left(\sum_{k=1}^{N}(N+1-k)x_{-k}\right) - \frac{N+1}{2}\left(\sum_{k=1}^{N}x_{-k}\right)}{N(N^2-1)/12}$$

$$\hat{a}_{-1} = \frac{1}{N}\left(\sum_{k=1}^{N}x_{-k}\right) - \frac{N+1}{2}\hat{b}_{-1}$$

Trend Seasonal

$$x_t = (a + bt)F_t + \varepsilon_t$$

1. Estimate seasonal indices.

2. Estimate trend line.

Trend Seasonal: seasonal indices

period t	demand x_t	$(x_{t-6}+x_{t-5}+\ldots+x_{t+5})/12$ (A_t)	$(x_{t-5}+x_{t-4}+\ldots+x_{t+6})/12$ (B_t)	$C_t=(A_t+B_t)/2$
-24 **a**	10			
-23 **a b**	14			
-22 **a b**	13			
-21 **a b**	20			
-20 **a b**	30			
-19 **a b**	33		21.0	
-18 **a b**	32	21.0 **a**	21.6 **b**	21.3 **(a+b)/2**
-17 **a b**	20	21.6	22.3	21.9
-16 **a b**	26	22.3	23.2	22.7
-15 **a b**	19	23.2	23.8	23.5
-14 **a b**	18	23.8	23.6	23.7
-13 **a b**	17	23.6	24.5	24.0
-12 **b**	17	24.5	25.6	25.0
-11	22	25.6	27.1	26.3

Trend Seasonal: seasonal indices

period t	demand x_t	C_t	ratio $R_t = x_t / C_t$	Index $F_t = 12R_t / Sum$
-19	33			
-18	32	21.3	1.50	1.56
-17	20	21.9	0.91	0.95
-16	26	22.7	1.14	1.19
-15	19	23.5	0.81	0.84
-14	18	23.7	0.76	0.79
-13	17	24.0	0.71	0.73
-12	17	25.0	0.68	0.71
-11	22	26.3	0.84	0.87
-10	24	27.6	0.87	0.90
-9	28	28.7	0.98	1.01
-8	27	29.6	0.91	0.95
-7	44	30.5	1.44	1.50
-6	45		Sum=11.55	Sum=12

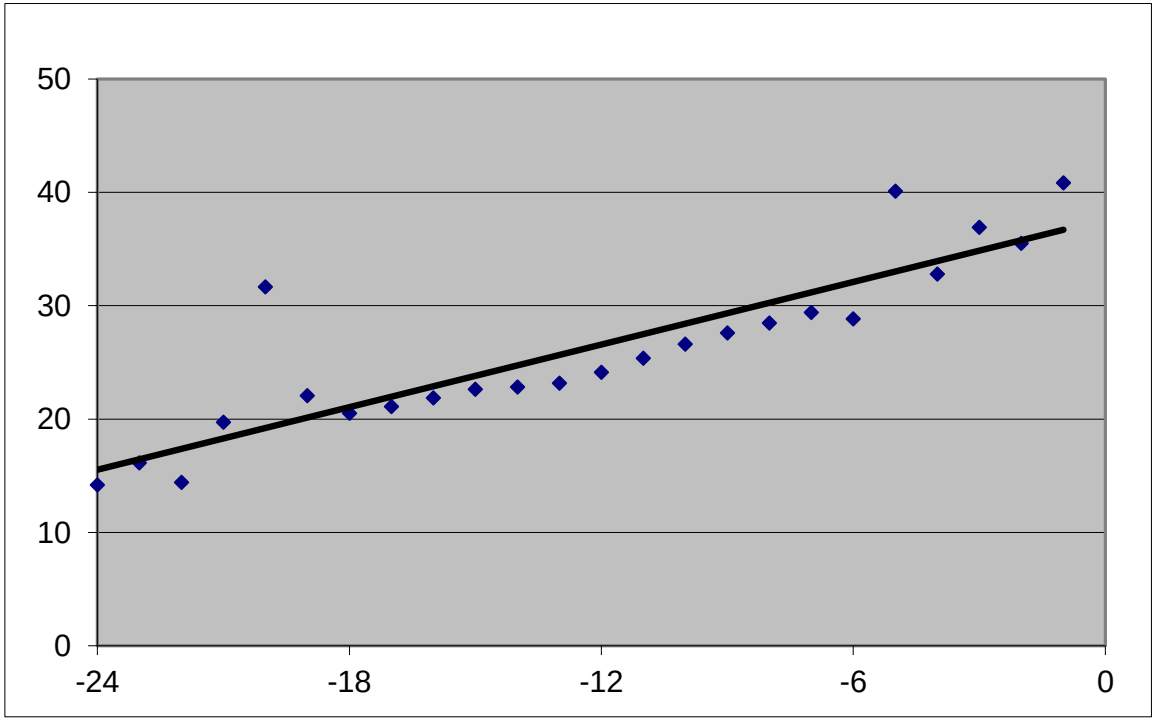
Note 1: The indices are copied to the other periods. E.g. $F_{-19}=F_{-7}=1.50$

Note 2: Demand history for at least two whole seasons is needed!

Trend Seasonal: trend line

Regression on estimate of level x_t / F_t

estimate of level x_t / F_t



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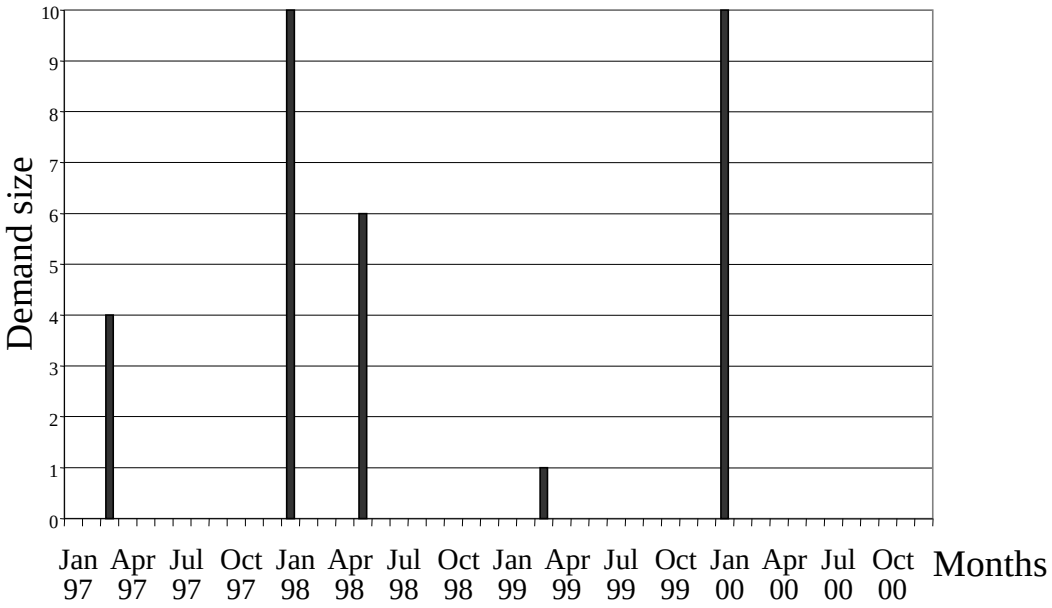
Intermittent demand



Any Stock Keeping Unit (SKU) within the range of products at any level of a given supply chain

- May account for up to 60% of the total stock value in any industrial setting

$\Rightarrow$ Small improvements imply substantial cost savings

Croston's Method - Motivation

- John Croston – Statistician at P&O, London.
- Audited stock control system, which used Exponential Smoothing for all low demand items.
- For some low demand items, stock levels were excessive. Errors were associated with intermittent and lumpy demand items.

Croston's Method (1972)

- If a (positive) demand occurs in period t, then update
  - Demand interval: $\hat{k}_t = (1-\alpha)\hat{k}_{t-1} + \alpha k_t$
  - Demand size: $\hat{d}_t = (1-\beta)\hat{d}_{t-1} + \beta x_t$
  - Demand per period: $\hat{a}_t = \hat{d}_t / \hat{k}_t$

- Otherwise
  - Do nothing

Croston: example ($\alpha=\beta=0.2$)

			4								
10				6							
		1									
10											

Initialize using say first two years gives:

- An initial interval estimate of 24/3 = 8 months
- An initial size estimate of (4+10+6)/3 = 6.7

Update after demand of size 1:

- 0.8*8 + 0.2*10 = 8.4 months
- 0.8*6.7 + 0.2*1 = 5.5

Update after demand of size 10:

- 0.8*8.4 + 0.2*10 = 8.6 months
- 0.8*5.5 + 0.2*10 = 6.4

NOTE for Croston

- They estimate the probability and mean size of positive demand **per period**.
- However, since there can be more than one customer in a period, these are NOT unbiased estimates of the arrival rate and the mean demand size **per customer**!

Compound Poisson demand

- Croston estimates the average demand per period and the time between periods with positive demand

- How do we estimate the average demand per customer (μ_c) and the demand rate (λ)?
- That is, how do we estimate the parameters of a compound Poisson demand process?

Prak et al. (2019)

- Assume that we have demand data for *n* periods
- Let N_0 be the stochastic number of periods with zero demand and n_o its realisation
- The probability of zero demand is $e^{-\lambda}$ and so $E[N_0] = n \times e^{-\lambda}$
- We set this equal to the sample moment n_o and get $e^{-\lambda} = n_o / n$,
- Estimator: $\hat{\lambda} = -\ln(n_o / n)$

Prak et al. (2019) cont'd

- And equate (as with standard MM) central moments of the compound Poisson distribution to their sample equivalents

- First order: mean demand (μ_1) per period equal to mean observed demand ($\hat{\mu}_1$)
- Since $\mu_1 = \lambda\mu_c$we solve $\hat{\mu}_1 = \lambda\hat{\mu}_c$
- To get $\hat{\mu}_c = \hat{\mu}_1/\hat{\lambda}$

$\wedge$

Previous example

			4								
10				6							
		1									
10											

Recall Croston Initialization:

- An initial interval estimate of 24/3 = 8 months
- An initial size estimate of (4+10+6)/3 = 6.7

We have: $n = 24, n_0 = 21, \hat{\mu}_1 = \frac{4+10+6}{24} = \frac{10}{12}$

- So we get $\hat{\lambda} = -\ln\left(\frac{n_0}{n}\right) = 0.134 \quad (> 1/8!)$
- And $\hat{\mu}_c = \frac{\hat{\mu}_1}{\hat{\lambda}} = 6.24 \quad (< 6.7!)$

New example

8	4		4		8	14	6		11	20	2
10	10		5	6		8		22	5	6	11

Croston Initialization would give:

- An initial interval estimate of 24/18 = 1.3 months
- An initial size estimate of (8+4+…+11)/18 = 160/18=8.9

We have: $n = 24, n_0 = 6, \hat{\mu}_1 = \frac{160}{24} = 6.7$

- So we get $\hat{\lambda} = -\ln\left(\frac{n_0}{n}\right) = 1.39 \quad (\gg 1/1.3!)$
- And $\hat{\mu}_c = \frac{\hat{\mu}_1}{\hat{\lambda}} = 4.8 \quad (\ll 8.9!)$

Forecast errors

Needed for inventory control

MAD and MSE - Example

Period	Demand	Forecast MA(3)	AD	SE
1	126,9			
2	161,5			
3	71,9			
4	133,7	120,1	13,6	185,0
5	97,9	122,4	24,4667	598,6
6	100,5	101,2	0,66667	0,4
7	78,4	110,7	32,3	1043,3
8	121,6	92,3	29,3333	860,4
9	66,1	100,2	34,0667	1160,5
10	104,2	88,7	15,5	240,3

MAD=21,4 MSE=584,1

Let X be a stochastic variable with mean m = E(X).

$\sigma = \sqrt{E(X-m)^2}$ and this can be directly estimated by $\sqrt{MSE}$

However, if MAD is used (e.g. to reduce the effect of outliers), then this is an estimate of E |X-m| and not of σ.

If X is normally distributed then $\sigma=\sqrt{(\pi/2)}$ E[MAD]≈1.25 E[MAD].

Based on this, we can estimate σ by 1.25 MAD.
Do keep in mind that this may not be accurate for other distributions.

Approximate st.d. per period

- Use either $\sqrt{MSE}$ or $1.25 \times MAD$

- Example: 24.2 or 26.8

Other factors

- Events (e.g. Easter)
- Price changes
- sales campaigns
- conflicts that affect demand
- new products without historical data
- new competitive products on the market
- new regulations